WORKED SOLUTION

STEP II 1999, Question 12

Probability and Statistics • Micro-chip

STEP II 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q12

The key idea

This is Bayes' theorem with a parameter left in. Once $\mathrm{P}(S)$ is found by the law of total probability, both conditional probabilities are simple rational functions of p, and each is monotonic on the allowed interval. So the "greatest value" in each inspector's procedure sits at an end of the interval $0 \le p \le \frac{1}{3}$, and the interesting part is what that says about estimating from a single observation.

Solution

The conditional probabilities

Let A, B, C be the events that a randomly selected chip was made by A, B, C respectively, and let S be the event that it is sound. These three events partition the sample space, with

$$\begin{aligned} \mathrm{P}(A) = 2p, \quad \mathrm{P}(B) = p, \quad \mathrm{P}(C) = 1 - 3p, \\ \mathrm{P}(S \mid A) = \tfrac{7}{10}, \quad \mathrm{P}(S \mid B) = \tfrac{8}{10}, \quad \mathrm{P}(S \mid C) = \tfrac{9}{10}. \end{aligned}$$

By the law of total probability,

$$\begin{aligned} \mathrm{P}(S) &= 2p \cdot \tfrac{7}{10} + p \cdot \tfrac{8}{10} + (1 - 3p) \cdot \tfrac{9}{10} \\ &= \tfrac{1}{10}(14p + 8p + 9 - 27p) = \tfrac{1}{10}(9 - 5p). \end{aligned}$$

Note that $9 - 5p > 0$ throughout $0 \le p \le \frac{1}{3}$, so we may divide by it. By Bayes' theorem,

$$\mathrm{P}(A \mid S) = \frac{\mathrm{P}(S \mid A)\,\mathrm{P}(A)}{\mathrm{P}(S)} = \frac{\frac{14p}{10}}{\frac{9-5p}{10}} = \frac{14p}{9 - 5p},$$

and similarly

$$\mathrm{P}(C \mid S) = \frac{\frac{9}{10}(1 - 3p)}{\frac{1}{10}(9 - 5p)} = \frac{9(1 - 3p)}{9 - 5p} = \frac{9 - 27p}{9 - 5p}.$$

The first inspector's estimate

As p increases from 0 to $\frac{1}{3}$, the numerator $14p$ increases and the positive denominator $9-5p$ decreases, so $\mathrm{P}(A \mid S)$ is strictly increasing. (Equivalently, its derivative is $126/(9-5p)^2 > 0$.) Its greatest value is therefore at the right-hand end, $p = \frac{1}{3}$, where it equals

$$\frac{14/3}{9-5/3} = \frac{14/3}{22/3} = \frac{7}{11}.$$

So the first inspector's estimate is $\hat{p} = \frac{1}{3}$.

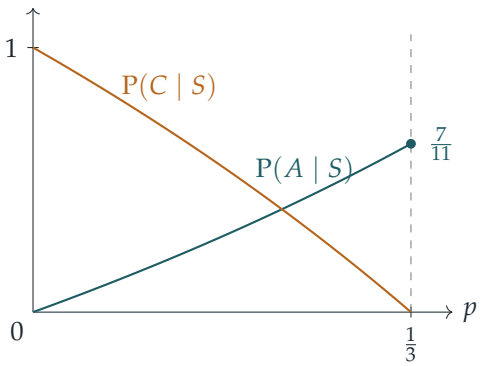
The second inspector's estimate

> **Ask yourself** Is $\mathrm{P}(C \mid S)$ increasing or decreasing? Both its numerator and its denominator decrease as p grows.

When numerator and denominator move in the same direction we can't just read off the answer, so we differentiate with the quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}p}\frac{9-27p}{9-5p} = \frac{-27(9-5p)+5(9-27p)}{(9-5p)^2} = \frac{-243+135p+45-135p}{(9-5p)^2} = \frac{-198}{(9-5p)^2} < 0.$$

So $\mathrm{P}(C \mid S)$ is strictly decreasing on $[0, \frac{1}{3}]$, and its greatest value is at $p = 0$, where it equals 1. The second inspector's estimate is $\hat{p} = 0$.



Comment

The two inspectors, using the same method on equally valid single observations, arrive at the two opposite extremes of the allowed range: $\hat{p} = \frac{1}{3}$ (no chips at all come from C) and $\hat{p} = 0$ (every chip comes from C). Each estimate is simply "the value of p that makes my chip's maker as common as possible", which a sample of one is bound to produce. The second estimate is even contradicted by the first inspector's chip, since $p = 0$ says that A makes no chips. Neither estimate is believable: one observation carries far too little information to estimate p, and a larger sample (with the counts from each manufacturer) is needed.

It is also worth noticing that the soundness of the chip plays almost no part. The likelihood of "sound and from A" is $\frac{7}{10} \cdot 2p$, which is maximised at the same place as $2p$ itself; the factor $\frac{7}{10}$ doesn't depend on p.

Checks

- At $p = 0$ every chip comes from C, so $\mathrm{P}(C \mid S)$ should be 1 and $\mathrm{P}(A \mid S)$ should be 0; both formulae agree.
- Similarly $\mathrm{P}(B \mid S) = 8p/(9 - 5p)$, and the three conditional probabilities add to
$$\frac{14p + 8p + 9 - 27p}{9 - 5p} = 1.$$
- At $p = 0.2$ the formulae give $\mathrm{P}(A \mid S) = 0.35$ and $\mathrm{P}(C \mid S) = 0.45$; a simulation of 400 000 chips gave 0.3505 and 0.4505.

Common slips

- Setting the derivative equal to zero and looking for a turning point. There isn't one: both maxima are at endpoints of $[0, \frac{1}{3}]$.
- Forgetting that $\mathrm{P}(C \mid S)$ needs a careful argument: both numerator and denominator decrease, so "the numerator gets smaller" is not enough.

Beyond STEP

Both inspectors are doing, in effect, maximum likelihood estimation: choosing the parameter value under which what they saw is most probable. With one observation the maximum likelihood estimate often lands on the boundary of the parameter space, which is a warning that it needs plenty of data to behave well. A Bayesian would instead put a prior distribution on p and update it with each chip, which avoids such extreme conclusions from a single observation. Blitzstein and Hwang, *Introduction to Probability*, discusses both points of view.

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